

ECOM30001/90001 Basic Econometrics

Tutorial 4: Testing Hypotheses about Regression Parameters

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Tutorial 4 Overview & Key Concepts

Core Topics

- **Single Hypothesis Testing (t -test):** One-tailed vs. two-tailed tests, linear combinations of parameters.
- **Joint Hypothesis Testing (F -test):** Testing overall significance ($\beta_1 = \beta_2 = 0$) and structural constraints.
- **Model Fit & Goodness of Fit:** Calculating R^2 and $\bar{R}^2$ from TSS and RSS .
- **Log-Log Production Models:** Elasticity interpretations and testing Cobb-Douglas restrictions ($\beta_1 = -1, \beta_2 = 0$).
- **R Practice:** Implementing t and F tests using `'car::linearHypothesis()'`.

Question 1(a): Goodness of Fit (R^2 and $\bar{R}^2$)

Model Setup & Given Information

Model: $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$ ($n = 20, k = 2$)

Estimated variance $\hat{\sigma}^2 = 2.5193$, sample standard deviation $s_Y = 6.4971$.

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Hint:

- **Sample Variance & TSS:** $s_Y^2 = \frac{TSS}{n-1} \implies TSS = (n-1)s_Y^2$
- **Error Variance & RSS:** $\hat{\sigma}^2 = \frac{RSS}{n-k-1} \implies RSS = (n-k-1)\hat{\sigma}^2$
- **R^2 and $\bar{R}^2$ Formulas:**

$$R^2 = 1 - \frac{RSS}{TSS}, \quad \bar{R}^2 = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)}$$

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Step-by-Step Derivation:

- **Total Sum of Squares (TSS):**

$$s_Y^2 = \frac{TSS}{n-1} \implies TSS = (n-1)s_Y^2 = (19)(6.4971^2) = \mathbf{802.034}$$

- **Residual Sum of Squares (RSS):**

$$\hat{\sigma}^2 = \frac{RSS}{n-k-1} \implies RSS = (n-k-1)\hat{\sigma}^2 = (17)(2.5193) = \mathbf{42.828}$$

- **Coefficient of Determination (R^2):**

$$R^2 = 1 - \frac{RSS}{TSS} = 1 - \frac{42.828}{802.034} = \mathbf{0.9466} \quad (94.66\%)$$

- **Adjusted R^2 ($\bar{R}^2$):**

$$\bar{R}^2 = 1 - \frac{RSS/(n-k-1)}{TSS/(n-1)} = 1 - \frac{42.828/17}{802.034/19} = \mathbf{0.9403} \quad (94.03\%)$$

Question 1(b): One-Sided t -Test on β_1

Hypotheses

$H_0 : \beta_1 \geq 1$ vs. $H_1 : \beta_1 < 1$ (Lower-tail test at $\alpha = 0.05$)

Question 1(b): One-Sided t -Test on β_1

Hypotheses

$H_0 : \beta_1 \geq 1$ vs. $H_1 : \beta_1 < 1$ (Lower-tail test at $\alpha = 0.05$)

Calculations & Decision:

- Given: $b_1 = 0.69914$,
 $\widehat{\text{var}}(b_1) = 0.048526 \implies \text{se}(b_1) = \sqrt{0.048526} = 0.220286$.

- Test Statistic:

$$t = \frac{b_1 - 1}{\text{se}(b_1)} = \frac{0.69914 - 1}{0.220286} = -1.3658 \sim t_{17}$$

- p -value from R: 'pt(-1.3658, df = 17)' = 0.0949.

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Conclusion

Since $p = 0.0949 > 0.05$, we **do not reject** H_0 at the 5% level. There is insufficient sample evidence to conclude that $\beta_1 < 1$.

Question 1(c): Joint F -Test ($\beta_1 = \beta_2 = 0$)

Hypotheses

$H_0 : \beta_1 = \beta_2 = 0$ vs. $H_1 : \text{at least one } \beta_j \neq 0$

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Hypotheses

$H_0 : \beta_1 = \beta_2 = 0$ vs. $H_1 : \text{at least one } \beta_j \neq 0$

F-Statistic Formula & Calculation:

- Under H_0 , restricted model is $Y_i = \beta_0 + \varepsilon_i \implies RRSS = TSS = 802.034$.
- Unrestricted model $URSS = 42.8281$ with $q = 2$ restrictions and $n - k - 1 = 17$ df.

$$F = \frac{(RRSS - URSS)/q}{URSS/(n - k - 1)} = \frac{(802.034 - 42.8281)/2}{42.8281/17} = \mathbf{150.678} \sim F_{2,17}$$

- **p-value from R:** 'pf(150.678, 2, 17, lower.tail = FALSE)' = $\mathbf{1.53 \times 10^{-11}}$.

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Conclusion

Since $p < 0.05$, we **reject** H_0 decisively at the 5% level. The regressors X_1 and X_2 are jointly statistically significant.

Question 1(d): Two-Sided Test on $\beta_2 = 2\beta_1$

Hypotheses

$$H_0 : \beta_2 - 2\beta_1 = 0 \quad \text{vs.} \quad H_1 : \beta_2 - 2\beta_1 \neq 0 \quad (\alpha = 0.05)$$

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Standard Error & t -Statistic:

- **Point Estimate:** $b_2 - 2b_1 = 1.7769 - 2(0.69914) = \mathbf{0.37862}$.

- **Variance of Linear Combination:**

$$\begin{aligned}\text{var}(b_2 - 2b_1) &= \text{var}(b_2) + 4\text{var}(b_1) - 4\text{cov}(b_2, b_1) \\ &= 0.037120 + 4(0.048526) - 4(-0.031223) = \mathbf{0.356116}\end{aligned}$$

$$\implies \text{se}(b_2 - 2b_1) = \sqrt{0.356116} = \mathbf{0.59675}$$

- **t -statistic:**

$$t = \frac{0.37862 - 0}{0.59675} = \mathbf{0.6345} \implies p\text{-value} = 2 \times \text{pt}(-0.6345, 17) = \mathbf{0.5342}.$$

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Conclusion

Since $p = 0.5342 > 0.05$, we **do not reject** H_0 at the 5% level.

Question 2(a)-(c): Log-Log Production Model

Econometric Model

$$\ln \left(\frac{L_t}{K_t} \right) = \beta_0 + \beta_1 \ln \left(\frac{w_t}{r_t} \right) + \beta_2 \ln Y_t + \varepsilon_t$$

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Interpretations & OLS Hypothesis Tests:

- **Economic Elasticities:** β_1 is the price elasticity of L/K ; β_2 is the output elasticity of L/K .

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Interpretations & OLS Hypothesis Tests:

- **Economic Elasticities:** β_1 is the price elasticity of L/K ; β_2 is the output elasticity of L/K .
- **Part (b) Test w/r Negative Effect ($H_0 : \beta_1 \geq 0$ vs $H_1 : \beta_1 < 0$):**
 $\hat{\beta}_1 = -0.58018$, two-sided $p = 8.78 \times 10^{-9}$.
One-sided p -value = $p/2 = 4.39 \times 10^{-9} < 0.05 \implies$ **Reject H_0 .** Relative price ratio has a statistically significant negative effect on L/K .

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One-sided p -value = $p/2 = 4.39 \times 10^{-9} < 0.05 \implies$ **Reject H_0** . Relative price ratio has a statistically significant negative effect on L/K .
- **Part (c) Test Output Effect ($H_0 : \beta_2 = 0$ vs $H_1 : \beta_2 \neq 0$):** $\hat{\beta}_2 = -0.23792$, $p = 0.000066 < 0.05 \implies$ **Reject H_0** . Output significantly affects L/K .

Question 2(d)-(e): Goodness of Fit & Joint Test in

Part (d): Proportion of Variation Explained

Multiple $R^2 = 0.9300 \implies 93\%$ of the variation in $\ln(L/K)$ is explained by movements in $\ln(w/r)$ and $\ln Y$.

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Part (d): Proportion of Variation Explained

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Part (e): Joint Hypothesis Test ($H_0 : \beta_1 = \beta_2 = 0$):

- **R Command:** `'linearHypothesis(mod, c("log(w/r)=0", "log(Y)=0"))'`
- **Console Output:**

$$F = 192.69, \quad p\text{-value} < 2.2 \times 10^{-16}$$

- **Conclusion:** Decisively **reject** H_0 at 5% level ($p < 0.05$). Relative factor price and output jointly affect the L/K ratio.

Question 2(f): Testing Cobb-Douglas Production Fu

Cobb-Douglas Restricted Model

Restricted Model: $\ln(L_t/K_t) = \beta_0 - \ln(w_t/r_t) + \varepsilon_t \implies$ Restrictions:
 $\beta_1 = -1$ and $\beta_2 = 0$.

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Hypothesis Testing & Manual F -Replication:

- **R Output ('linearHypothesis'):** $F = 16.616$,
 $p\text{-value} = 0.0000155 < 0.05 \implies$ **Reject Cobb-Douglas restrictions** at 5% level.
- **Manual F -statistic Replication:** $RRSS = 1.56306$, $URSS = 0.72838$, $q = 2$,
 $n - k - 1 = 32 - 2 - 1 = 29$.

$$F = \frac{(RRSS - URSS)/q}{URSS/(n - k - 1)} = \frac{(1.56306 - 0.72838)/2}{0.72838/29} = \mathbf{16.616}$$

Matches R output perfectly!

Exam Practice Question (MCQ)

Question Setup

Fit $Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$ with $n = 43$, overall $F = 100$, and $RSS = 20$. Find Total Sum of Squares (TSS).

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Step-by-Step Solution:

- Overall F -Statistic Formula:

$$F = \frac{(TSS - RSS)/k}{RSS/(n - k - 1)} \quad \text{where } k = 2, n - k - 1 = 40$$

- Substitute Values:

$$100 = \frac{(TSS - 20)/2}{20/40} = \frac{(TSS - 20)/2}{0.5} = TSS - 20$$

- Solve for TSS : $100 = TSS - 20 \implies TSS = 120$.

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$$100 = \frac{(TSS - 20)/2}{20/40} = \frac{(TSS - 20)/2}{0.5} = TSS - 20$$

- Solve for TSS : $100 = TSS - 20 \implies TSS = 120$.

Correct Answer

Option (2) 120